The Theory of Automata & Formal Languages

AI412

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DFA for strings containing 01:



- Q = {q_0, q_1, q_2}
- ∑ = {0,1}
- start state = q0
- F = {q_2}
- Transition table.

	symbols	
states / δ	0	1
q_0	q_1	q_0
q_1	q_1	q_2
*q_2	q_2	q_2

Worked Example: Show that the language: L: {awa: w ϵ {a, b} *} is regular.

Solution:

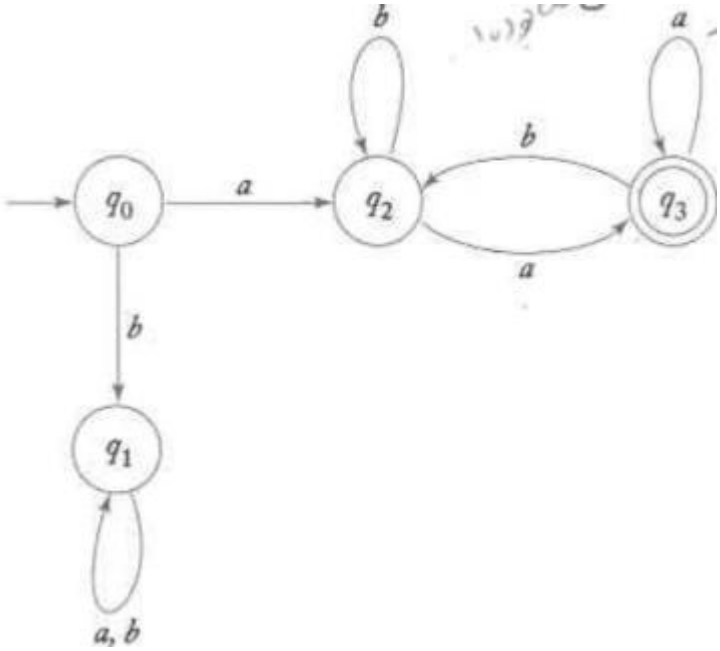
1. Since w ϵ {a, b} *, then empty string with us. So, the first string we can construct the DFA with it "aa " under condition that at the end of any testing string the movement is at the final/accept state.

2. Now, test with many examples like: aaa, aba, aaba, abaa, aabababa, … so on.

3. You must note that the alphabet contains 2 symbols,

so you must aware to make any state give 2 transitions (2 reads: one for "a" another for "b").

4. Repeat to finalize all testing probabilities.



Worked Example: Determine the DFA schematic for M = (Q, Σ , δ, q, F) where Q = {q_1, q_2, q_3}, Σ = {0,1}, q_1 is the start state, F = {q_2} and δ is given by the table below:

Initial state q	Symbol σ	Final state $\delta(q,\sigma)$
q_1	0	q_1
q_1	1	q_2
q_2	0	q_3
q_2	1	q_2
q_3	0	q_2
q_3	1	q_2

Also determine a Language L recognized by the DFA.

Solution:

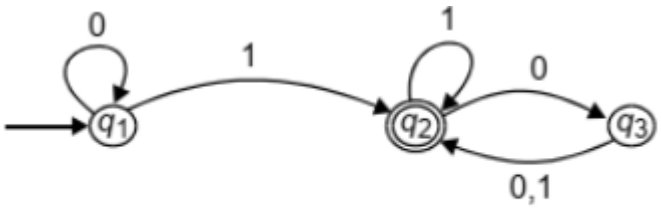


Fig. Finite Automaton having three states.

From the given table for δ, the DFA is drawn, where q2 is the only final state. (It is to be noted that a DFA can “accept” a string and it can “recognize” a language. Catch here is that “accept” is used for strings and “recognize” for that of a language).

It could be seen that the DFA accepts strings that has at least one 1 and an even number of 0s following the last 1. Hence the language L is given by:

L = {w | w contains at least one 1 and an even number of 0s follow the last 1}.

Worked Example: Sketch the DFA given M = ({q_1, q_2}, {0 ,1}, δ, q_1, {q_2}) and δ is given by:

$$\delta (q_1 ,0) = q_1, \text{ and } \delta (q_2 ,0) = q_1$$

$$\delta (q_1 ,1) = q_2, \delta (q_2 ,1) = q_2$$

Determine a Language L(M), that the DFA recognizes.

Solution: From the given data, it is easy to predict the schematic of DFA as follows:

Internal states = q_1, q_2.

Symbols = 0, 1.

Transition function = δ

q_1 = Initial state

q_2 = Final state.

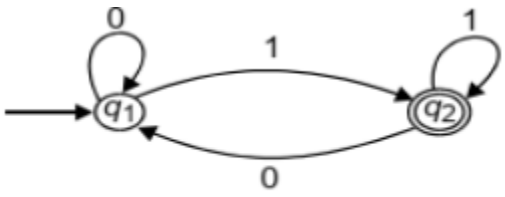


Fig. State diagram of DFA

If a string ends in a 0, it is “rejected” and “accepted” only if the string ends in a 1.

Therefore, the language: L(M) = {w | w ends in a 1}.

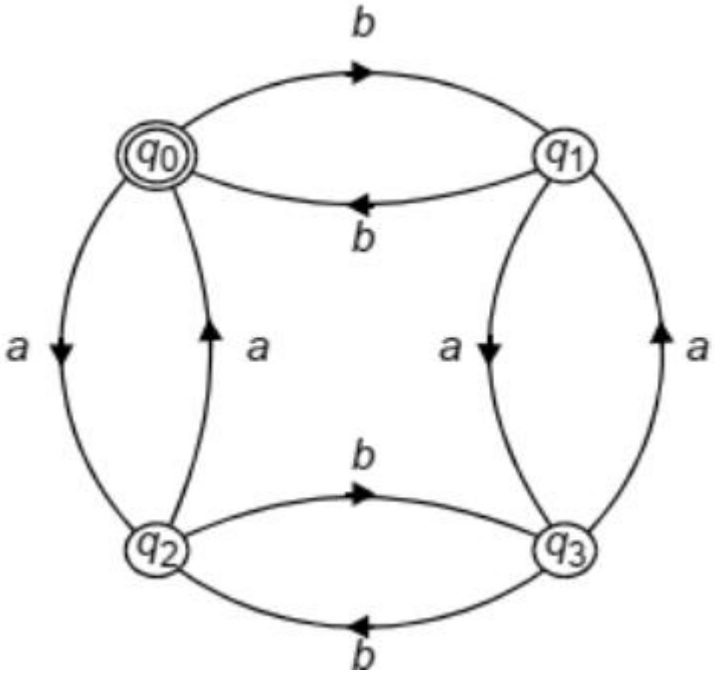
Worked Example: Obtain the state table diagram and state transition diagram (DFA Schematic) of the finite state Automaton M = (Q, Σ, δ, q_0, F), where Q = {q_0, q_1, q_2, q_3}, Σ = {a, b}, q_0 is the initial state, F is the final state with the transition defined by:

$$\begin{array}{lll} \delta(q_0,a)=q_2 & \delta(q_3,a)=q_1 & \delta(q_2,b)=q_3 \\ \delta(q_1,a)=q_3 & \delta(q_0,b)=q_1 & \delta(q_3,b)=q_2 \\ \delta(q_2,a)=q_0 & \delta(q_1,b)=q_0 & \end{array}$$

The State Table diagram is as shown:

δ	a	b
q_0	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

Solution: With the given definitions, the State Transition diagram/DFA Schematic is



Unit 3: Nondeterministic Finite Automata (NFA) and Equivalence to DFA:

This unit introduces nondeterminism in finite automata and proves, via subset construction, that NFAs and DFAs recognize exactly the same languages.

Historical Note: Nondeterminism as a formal modeling tool was introduced by Rabin and Scott in their influential 1959 paper, which also proved the DFA-NFA equivalence and later earned them the Turing Award in 1976.

A Nondeterministic Finite Automata (NFA) is defined by a 5-tuple

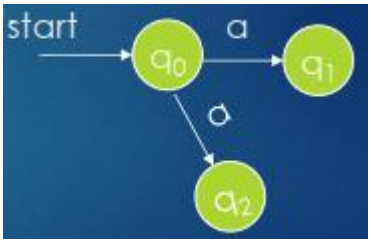
M = (Q, Σ, δ, q0, F) where Q, Σ, δ, q_0, F are defined as follows:

Q = Finite set of internal states
Σ = Finite set of symbols called "Input alphabet"
$\delta = Q \times (\Sigma \cup \{\lambda\}) \rightarrow 2^Q$
$q_0 \in Q$ is the Initial states
$F \subseteq Q$ is a set of Final states

NFA differs from DFA in that, the range of δ in NFA is in the powerset 2^Q. A string is accepted by an NFA if there is some sequence of possible moves that will put the machine in the final state at the end of the string.

Properties of NFA:

1. It is possible to make transition from state to another with the same reading (Ex: in Σ = {a , b}, we can move from q0 to q1 if we read a, also we can move from q0 to q2 by reading a).



2. It is possible to move from state to another without reading any symbol.



3. In NFA, you can move from state to other states in number of arrows smaller than the symbols in alphabet, in DFA you must be equal.

4. It is not necessary to go back in automaton; we always go forward.

5. It is not necessary to complete the string in reading, but we can stop when no path from the current state to new one.

6. The language in NFA is possibly come from many languages.

• The automaton in NFA like some of automata work in parallel.

• **Example:** Read: 010110



Solution:

